

human

Show me all the construction sites in Exeter.?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to see all the construction sites in Exeter. Provide the general steps to solve this query.'}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entity to search for is construction sites, and we need to find as many entries as possible for construction sites in Exeter.

output steps: ["Find Exeter 1 area", "Search for construction sites in Exeter as many search results"]

host_agent

Tool send_message returned result : The area is Exeter, so there will be 1 entry for Exeter. The entity to search for is construction sites, and we need to find as many entries as possible for construction sites in Exeter.

output steps: ["Find Exeter 1 area", "Search for construction sites in Exeter as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_polygon'} Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_polygon']} Tool generate_metadata_for_artifacts returned result : [...metadata...]

Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    filtered = gdf[(
        (gdf['name1_text'].str.lower() == 'exeter') |
        (gdf['name1_text'].str.lower().str.contains('exeter')) |
        (gdf['sameasdbpedia'].notnull() & gdf['sameasdbpedia'].str.lower().str.contains('exeter')) |
        (gdf['sameasgeonames'].notnull() & gdf['sameasgeonames'].str.lower().str.contains('exeter'))
    )]
    # Further restrict to relevant description groups for area polygons
    filtered = filtered[filtered['descriptiongroup'].isin(['Settlement', 'Part Of Settlement', 'City', 'Town', 'Suburban Area'])]
    summary = f"Filtered Exeter polygon(s): {len(filtered)} entries found."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'Polygon(s) representing the area of Exeter, filtered from the original dataset using name and external references.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered Exeter polygon(s): 2 entries found.',]

Filtered Exeter polygon(s): 2 entries found. An artifact has been generated with the name 'filtered_exeter_polygon' and contains polygon(s) representing the area of Exeter, filtered from the original dataset using name and external references. If you need to view or analyze these polygons further, let me know!

named_area_agent

I found the area polygon for Exeter. The filtered results are saved in the artifact named '**filtered_exeter_polygon**'. If you need further analysis or details, feel free to ask!

host_agent

Calling tool generate_metadata_for_all_artifacts with args : {} Tool generate_metadata_for_all_artifacts returned result : {'exeter_polygon': 'A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..', 'filtered_exeter_polygon': 'Polygon(s) representing the area of Exeter, filtered from the original dataset using name and external

references.}

Calling tool send_message with args : {'target': 'land_features', 'task_description': "Search for construction sites within the area defined by the artifact 'filtered_exeter_polygon'."}

land_features_agent

Calling tool call_os_ngd with args : {'filters': ['Construction Site'], 'bbox': 'filtered_exeter_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'construction_sites_exeter'} Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['land_construction_sites_exeter'],\n Descriptions are : ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for construction sites within the area defined by the artifact 'filtered_exeter_polygon'. using filters ['Construction Site']. "],\n counts of records fetched are : [84].\n ', []]

I found 84 construction sites within the area defined by the artifact 'filtered_exeter_polygon'. The data is saved in the artifact named 'land_construction_sites_exeter'.

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot all construction sites from 'land_construction_sites_exeter' within the area polygon 'filtered_exeter_polygon'."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['land_construction_sites_exeter', 'filtered_exeter_polygon']} Tool generate_metadata_for_artifacts returned result : [...metadata...]

Tool Code Executor

```
def plot_construction_sites_within_polygon(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon

    # Unpack data
    construction_sites = data[0].copy()
    exeter_polygons = data[1].copy()
```

```

# Ensure CRS is the same and set to EPSG:4326 for folium
if construction_sites.crs != exeter_polygons.crs:
    construction_sites = construction_sites.to_crs(exeter_polygons.crs)

# Use the largest polygon (likely the city boundary)
exeter_city_poly = exeter_polygons.loc[exeter_polygons['description'] == 'City']
if exeter_city_poly.empty:
    exeter_city_poly = exeter_polygons.iloc[[0]]

# Spatial join: keep only construction sites within the Exeter polygon
within_mask = construction_sites.within(exeter_city_poly.iloc[0].geometry)
construction_within = construction_sites[within_mask]

# Convert to EPSG:4326 for folium
construction_within = construction_within.to_crs(epsg=4326)
exeter_city_poly = exeter_city_poly.to_crs(epsg=4326)

# Create folium map centered on Exeter
centroid = exeter_city_poly.iloc[0].geometry.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Add Exeter polygon
folium.GeoJson(exeter_city_poly.iloc[0].geometry, name='Exeter City Boundary', style_function=lambda x: {'color': 'blue', 'fill': False, 'weight': 2}).add_to(m)

# Add construction sites
for _, row in construction_within.iterrows():
    folium.GeoJson(row.geometry, name='Construction Site', style_function=lambda x: {'color': 'red', 'fill': True, 'fillOpacity': 0.5, 'weight': 1}).add_to(m)

m.save('construction_sites_within_exeter.html')

summary = f"Plotted {len(construction_within)} construction sites within the Exeter city polygon. First 5 site IDs: {construction_within['osid'].head(5).to_list()}"
artifact_name = 'construction_sites_within_exeter_map'
artifact_description = 'A folium map showing all construction sites from the land_construction_sites_exeter dataset that are located within the Exeter city polygon'
artifact_data = 'construction_sites_within_exeter.html'
return [summary, artifact_name, artifact_description, artifact_data]

```

Tool code_executor returned result : ["Plotted 76 construction sites within the Exeter city polygon. First 5 site IDs: ['f9cfba5c-5055-4ee8-b825-21441c4d5527', '98dd6cd7-2c92-420f-9998-2526e31837bc', '00f10c5e-c8c4-4ea7-85e5-91e0a45057fb', 'd66aae4e-6a72-4979-b422-64b86a2bfc42', '363a7a65-6c5e-4368-b716-8e0ffbb92974']",]

Plotted 76 construction sites within the Exeter city polygon. First 5 site IDs: ['f9cfba5c-5055-4ee8-b825-21441c4d5527', '98dd6cd7-2c92-420f-9998-2526e31837bc', '00f10c5e-c8c4-4ea7-85e5-91e0a45057fb', 'd66aae4e-6a72-4979-b422-64b86a2bfc42', '363a7a65-6c5e-4368-b716-8e0ffbb92974']

Artifact name: construction_sites_within_exeter_map

Artifact description: A folium map showing all construction sites from the land_construction_sites_exeter dataset that are located within the Exeter city polygon.

Artifact data: construction_sites_within_exeter.html

host_agent

All construction sites within Exeter have been identified and plotted on a map. Here's what was done:

1. The area polygon for Exeter was found.
2. All construction sites within Exeter were searched and identified (84 found, 76 plotted).
3. A map was generated showing the locations of these construction sites within the Exeter city boundary.

If you would like to view or download the map, or need further details about any specific construction site, please let me know!